

Secure Wireless Networks

Wireless security configuration and unauthorized-access controls for the DeRexi corporate network

ASSIGNMENT OBJECTIVE

Document the security of one wireless network, including WPA2/WPA3 configuration and the WPS setting used to reduce unauthorized access.

Environment overview

DeRexi uses a corporate wireless design in which Dana's work laptop connects through a protected wireless segment before traffic reaches the access point, access switch, network firewall, router/gateway, and DeRexi application server. The existing design identifies WPA3 / 802.1X as the wireless security checkpoint.

Wireless security configuration

Wireless network / SSID	DeRexi-Secure (assignment configuration)
Security mode	WPA3-Enterprise
Network authentication	802.1X
WPS setting	Disabled
Authorized access	Approved DeRexi employees and managed devices
Corporate subnet	192.168.10.0/24
Default gateway	192.168.10.1
Example endpoint	Dana's laptop - 192.168.10.25/24

Why these controls are used

- WPA3-Enterprise protects wireless traffic in a business environment and is stronger than treating the network like a home Wi-Fi connection.
- 802.1X adds network-access authentication so a user or device must be authorized before joining the corporate network.
- WPS is intentionally disabled. This removes the push-button/PIN enrollment path and helps prevent unauthorized devices from bypassing the organization's normal authentication process.

IMPLEMENTATION NOTE

This report documents the DeRexi fictional lab/design environment. It does not claim a production router or access point was physically configured.

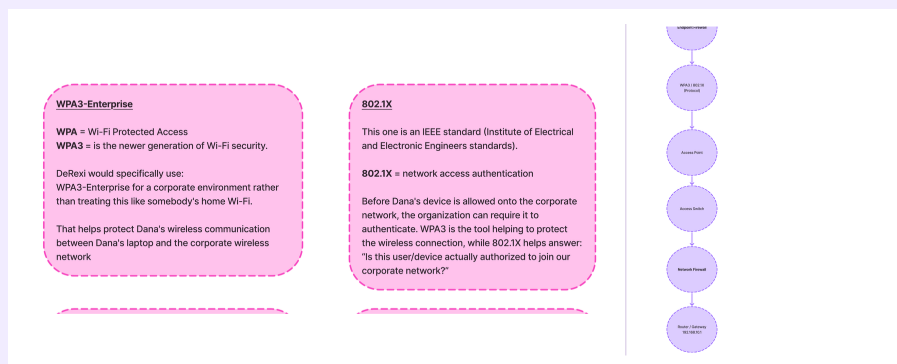
Implementation & Evidence

How the DeRexi wireless controls work together

Connection sequence

- 1 Dana selects the DeRexi corporate wireless network.
- 2 WPA3-Enterprise protects the wireless connection.
- 3 802.1X checks whether the user/device is authorized.
- 4 WPS remains disabled so easy-enrollment is not an alternate access path.
- 5 Authorized traffic continues through the AP, switch, firewall and gateway; HTTPS/TLS protects application traffic.

Evidence from the existing DeRexi network diagram



Unauthorized-access prevention

- **WPA3-Enterprise + 802.1X:** protect the wireless link and require authorized network access.
- **WPS disabled:** removes the PIN/push-button enrollment shortcut that could weaken the normal authentication process.
- **Layered controls:** authorized traffic still passes through the access point, switch, firewall, and gateway before reaching DeRexi resources.

RUBRIC COVERAGE CHECK

- ✓ One wireless network is documented.
- ✓ WPA3-Enterprise and 802.1X are documented as security/access-authentication controls.
- ✓ WPS is explicitly configured as disabled to reduce unauthorized enrollment/access risk.